

# SCE-012 — LLMs and Structural Cognitive Emergence

## Answer Generation vs Question Emergence

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*with chatGPT (OpenAI)*

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**Repository: <https://github.com/sizhet/Structural-Cognitive-Emergence-SCE>**

Modern Large Language Models (LLMs) have achieved remarkable success. They can:

- answer questions,
- write software,
- summarize documents,
- assist scientific research.

However, a central motivation behind SCE is the observation that:

Answer generation and question emergence are not the same problem.

Current LLMs excel at:

Prompt

↓

Answer

SCE focuses on:

Structure

↓

Gap

↓

Question

The framework was partly inspired by discussions raised by Transformer co-author Lukasz Kaiser concerning:

- human learning efficiency,
- long-context limitations,
- retrieval versus memorization,
- and the challenge of autonomous AI.

SCE does not argue against LLMs.

Instead, it explores a complementary question:

How do meaningful questions emerge?

Future cognitive systems may combine:

LLM

+

Structural Growth

+

Gap Detection

+

Question Emergence

to support more autonomous forms of intelligence.

In this view:

LLMs may provide a vast knowledge space.

Structural mechanisms may provide a growing question space.

Both may be required for future autonomous discovery systems.